EXPERIMENT NUMBER: 5

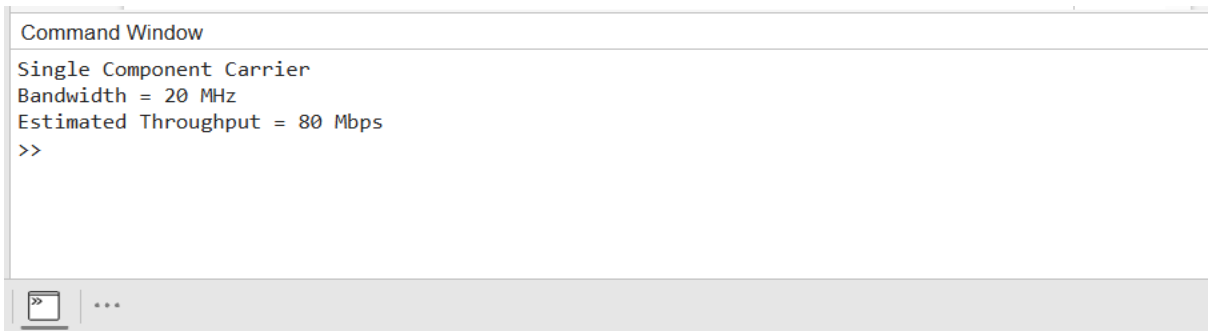
Simulation and Throughput Analysis of LTE-Advanced Carrier Aggregation

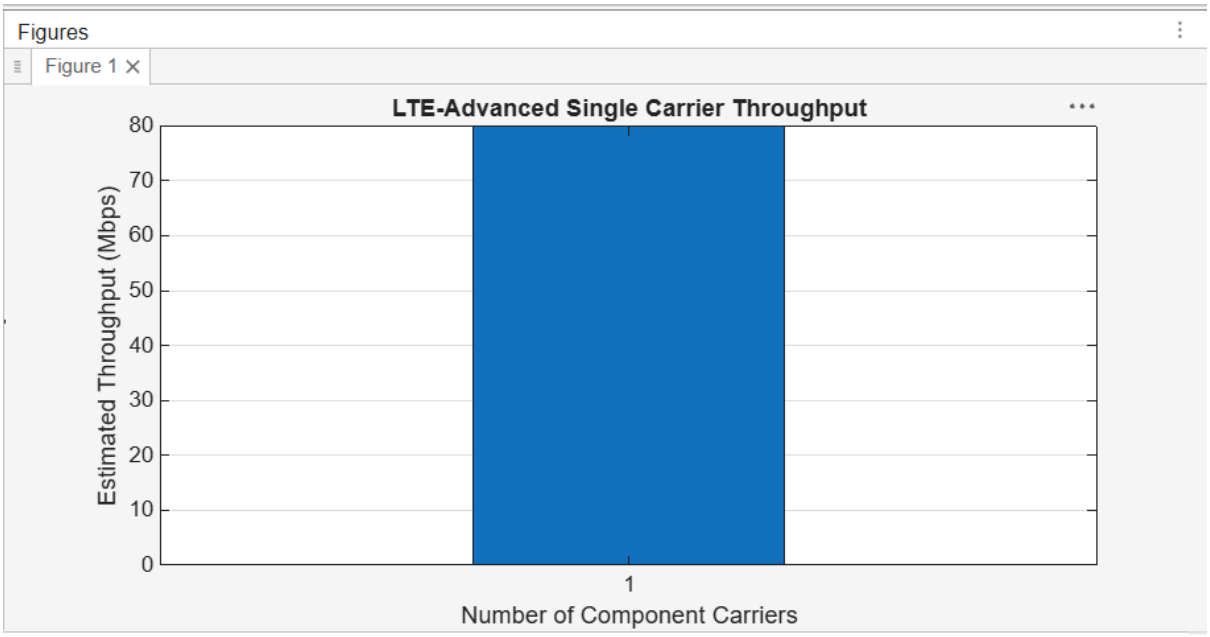
OBJECTIVE 1:

Matlab Program

```
clc;
clear;
close all;
BW1 = 20; % Component carrier bandwidth (MHz)
SE = 4; % Spectral efficiency (bits/s/Hz)
Ncc = 1; % Number of component carriers
BWagg = BW1*Ncc;
Throughput = BWagg*SE; % Mbps for MHz × bits/s/Hz
disp('Single Component Carrier')
disp(['Bandwidth = ',num2str(BWagg),' MHz'])
disp(['Estimated Throughput = ',num2str(Throughput),' Mbps'])
figure
bar(Ncc,Throughput)
grid on
xlabel('Number of Component Carriers')
ylabel('Estimated Throughput (Mbps)')
title('LTE-Advanced Single Carrier Throughput')
```

Result





OBJECTIVE 2:

Matlab Program

```
clc;
clear;
close all;
BWcc = 20; % MHz per component carrier
SE = 4; % bits/s/Hz
CC = [1 2 3]; % Number of component carriers
AggBW = BWcc .* CC; % MHz
Throughput = AggBW .* SE; % Mbps
disp('CC Aggregated BW (MHz) Estimated Throughput (Mbps)')
disp([CC' AggBW' Throughput'])
figure
bar(CC,Throughput)
grid on
xlabel('Number of Aggregated Component Carriers')
```

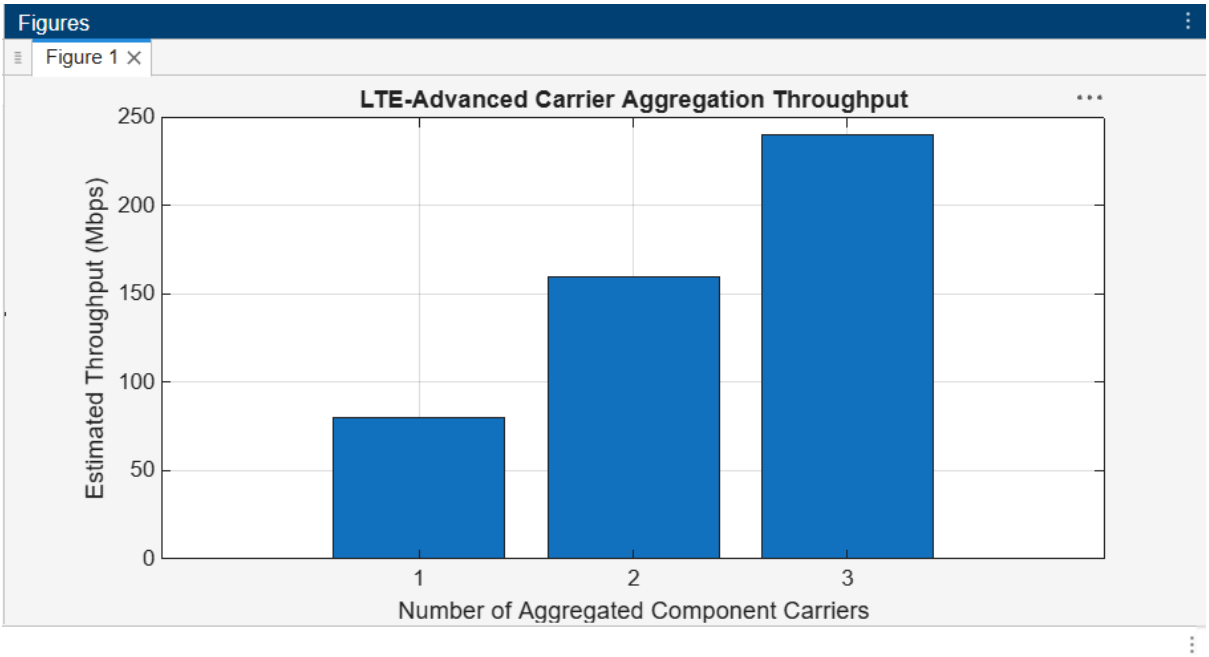
ylabel('Estimated Throughput (Mbps)')

title('LTE-Advanced Carrier Aggregation Throughput')

Result

```
Command Window
CC Aggregated BW (MHz) Estimated Throughput (Mbps)
     1    20    80
     2    40   160
     3    60   240

>>
```



OBJECTIVE 3:

Matlab Program

clc;

clear;

close all;

BW = [5 10 15 20]; % Carrier bandwidths (MHz)

SE = 4; % bits/s/Hz

TP = BW .* SE; % Estimated throughput (Mbps)

disp('Bandwidth (MHz) Throughput (Mbps)')

```
disp([BW' TP'])
figure
plot(BW,TP,'o-','LineWidth',2)
grid on
xlabel('Carrier Bandwidth (MHz)')
ylabel('Estimated Throughput (Mbps)')
title('Throughput vs Component Carrier Bandwidth')
```

Result

```
Command Window
Bandwidth (MHz) Throughput (Mbps)
     5    20
    10    40
    15    60
    20    80

>>
```

